

Unifying Special Relativity and Quantum Mechanics

Lecture 1 — The Euclidean Group, its Lie Algebra, and Unitary Irreps of $SE(2)$

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Everything in the normal text and in the **yellow boxes** is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted in the original). Everything inside a **blue box** like this one is extra material that I added to make the notes easier to follow: motivations, worked details of skipped algebra, and remarks. The figures are redrawn versions of the hand sketches. Page markers like [p. 3] indicate where each original page begins.

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1 Goal of the course [p. 1]

Goal: find a quantum mechanical description of a *massive particle* compatible with the postulates of special relativity.

► We start from a *non-relativistic* theory. We introduce momentum operators $\vec{P}$ with components such that

$$[P^i, P^j] = 0.$$

The Hamiltonian is

$$H = \frac{|\vec{P}|^2}{2m} \quad \rightsquigarrow \quad [H, \vec{P}] = 0.$$

We get a complete description of the system if we consider the common eigenstates of $\vec{P}$ described by the eigenvalue equation

$$\vec{P}|\vec{p}\rangle = \vec{p}|\vec{p}\rangle, \quad \langle\vec{p}|\vec{q}\rangle = \delta(\vec{p} - \vec{q}), \quad \int d^3p |\vec{p}\rangle\langle\vec{p}| = \mathbb{I}.$$

The Hamiltonian gives

$$H|\vec{p}\rangle = \frac{|\vec{P}|^2}{2m}|\vec{p}\rangle = \frac{|\vec{p}|^2}{2m}|\vec{p}\rangle \quad \rightsquigarrow \quad \text{we know the evolution of the states.}$$

Added explanation (not in the original notes)

Because H commutes with all the P^i , the momentum eigenstates are simultaneously energy eigenstates. So once we know $|\vec{p}\rangle$ at $t = 0$, its time evolution is simply a phase, $e^{-iE_p t}|\vec{p}\rangle$ with $E_p = |\vec{p}|^2/2m$: the theory is completely solved. The point of the whole construction that follows is to understand this familiar result from a *group-theoretic* point of view (symmetry under translations and rotations), because that is the language that generalizes to special relativity, where the Euclidean group of space will be replaced by the Poincaré group of spacetime.

[p. 2] ► We would like to find a *different interpretation of the same equations* that can be generalized to the relativistic case.

2 The Euclidean group and its subgroups

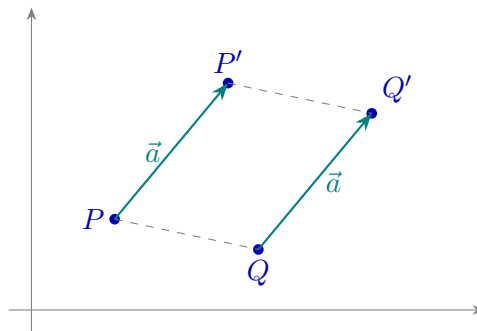
The Euclidean group $E(m)$ is the group of bijective *distance-preserving* maps of $\mathbb{R}^m$ to itself:

$$f : \mathbb{R}^m \rightarrow \mathbb{R}^m, \quad d(\vec{x}, \vec{y}) = d(f(\vec{x}), f(\vec{y})), \quad \text{with} \quad d(\vec{x}, \vec{y}) = |\vec{x} - \vec{y}|.$$

In more “colloquial” terms, the group contains translations, rotations, reflections, as well as combinations of all of them.

2.1 Example: $m = 2$

Translations. Translation by a vector $\vec{a}$ leaves the distance between two points P and Q invariant.



[p. 3] Two comments:

- (i) the distance *with respect to the origin* of P and Q *changes* after a translation;
- (ii) translations are non-linear transformations $\Leftrightarrow$ they do not verify a key property of linearity, which is vector addition:

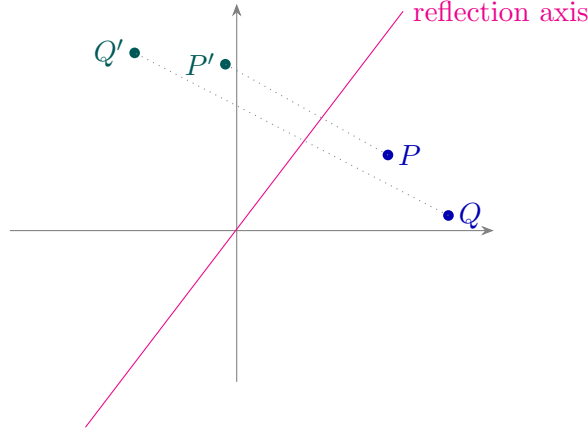
$$T(\vec{u} + \vec{v}) \neq T(\vec{u}) + T(\vec{v}).$$



Rotations. Rotation by θ leaves the distance between P and Q invariant. Two comments:

-

3



Rotations and reflections (w.r.t. a reflection axis that passes through the origin) are linear transformations that preserve the distance and, by themselves, they form a group: the orthogonal group $O(m)$.

2.2 The most general element of $E(m)$

We write the most general element of $E(m)$ in the form

$$\mathcal{E}[R(\vec{\theta}), \vec{a}]$$

where

- $\vec{a} \in \mathbb{R}^m$ identifies the translation $\rightsquigarrow m$ real parameters;
- $R(\vec{\theta}) \in O(m)$ identifies either a rotation or a reflection $\rightsquigarrow m(m-1)/2$ parameters.

[p. 5]

$$\dim(E(m)) = \frac{m(m+1)}{2} \quad \text{since} \quad m + \frac{m(m-1)}{2} = \frac{2m + m^2 - m}{2} = \frac{m(m+1)}{2}.$$

Added explanation (not in the original notes)

Where does the counting $m(m-1)/2$ for $O(m)$ come from? An orthogonal matrix satisfies $R^T R = \mathbb{I}$. A general real $m \times m$ matrix has m^2 entries; the condition $R^T R = \mathbb{I}$ is a symmetric matrix equation, i.e. $\frac{m(m+1)}{2}$ independent constraints. Hence the number of free parameters is $m^2 - \frac{m(m+1)}{2} = \frac{m(m-1)}{2}$. Check: for $m=2$ this gives 1 parameter (one rotation angle θ); for $m=3$ it gives 3 (the three Euler angles).

The *defining representation* is the representation in which the elements of $E(m)$ act on vectors $\vec{x} \in \mathbb{R}^m$. We write

$$E[R(\vec{\theta}), \vec{a}] \vec{x} = R(\vec{\theta}) \vec{x} + \vec{a}$$

where $R(\vec{\theta})$ is the element of $O(m)$ in the defining representation: it is an $m \times m$ real matrix such that $RR^T = R^T R = \mathbb{I}$.

E.g. for $m=2$:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \text{ for a rotation,} \quad R(\theta) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \text{ for a reflection w.r.t. the axis } x=0.$$

Added explanation (not in the original notes)

In the handwritten notes the rotation matrix appears with both off-diagonal signs written as $\sin \theta$; the standard (counter-clockwise) rotation matrix has $-\sin \theta$ in the upper-right entry, as written above. A quick sanity check: $\det R = \cos^2 \theta + \sin^2 \theta = +1$ for the rotation, while $\det = -1$ for the reflection — this sign of the determinant is exactly what will distinguish $SO(m)$ from $O_-(m)$ in Sec. 3.

2.3 Group properties of $E(m)$

Neutral element.

$$\mathcal{E}(\mathbb{K}, \vec{0})$$

“translation by nothing” ($\vec{0}$) together with the neutral element of $O(m)$ ($\mathbb{K}$).

Composition rule. [p. 6] To find out the composition rule, we consider the defining representation and write (with the short-hand notation $R(\vec{\theta}_2) = R_2$, $R(\vec{\theta}_1) = R_1$):

$$\begin{aligned} E[R_2, \vec{a}_2] E[R_1, \vec{a}_1] \vec{x} &= E(R_2, \vec{a}_2) (R_1 \vec{x} + \vec{a}_1) \\ &= R_2 (R_1 \vec{x} + \vec{a}_1) + \vec{a}_2 = R_2 R_1 \vec{x} + (R_2 \vec{a}_1 + \vec{a}_2) = \mathcal{E}(R_2 R_1, R_2 \vec{a}_1 + \vec{a}_2) \vec{x}. \end{aligned}$$

So we find

$$\mathcal{E}[R(\vec{\theta}_2), \vec{a}_2] \mathcal{E}[R(\vec{\theta}_1), \vec{a}_1] = \mathcal{E}[R(\vec{\theta}_2)R(\vec{\theta}_1), R(\vec{\theta}_2)\vec{a}_1 + \vec{a}_2].$$

Added explanation (not in the original notes)

Note the “twist” in the translation part: the result is *not* simply $\vec{a}_1 + \vec{a}_2$, but $R_2 \vec{a}_1 + \vec{a}_2$ — the second transformation rotates the first translation before adding its own. This twist is the algebraic fingerprint of the *semi-direct* product structure that will appear in Sec. 2.5. If the twist were absent, $E(m)$ would just be the direct product $T(m) \times O(m)$.

Inverse element. Given $\mathcal{E}(R, \vec{a})$, find $\mathcal{E}(\tilde{R}, \vec{b})$ such that $\mathcal{E}(R, \vec{a})\mathcal{E}(\tilde{R}, \vec{b}) = \mathcal{E}(\mathbb{K}, \vec{0})$. From the composition rule:

$$\mathcal{E}(R\tilde{R}, R\vec{b} + \vec{a}) = \mathcal{E}(\mathbb{K}, \vec{0}) \Rightarrow \begin{cases} R\tilde{R} = \mathbb{K} \Rightarrow \tilde{R} = R^{-1} \\ R\vec{b} + \vec{a} = \vec{0} \Rightarrow \vec{b} = -R^{-1}\vec{a} \end{cases}$$

[p. 7]

$$\mathcal{E}[R(\vec{\theta}), \vec{a}]^{-1} = \mathcal{E}[R(\vec{\theta})^{-1}, -R(\vec{\theta})^{-1}\vec{a}].$$

2.4 Subgroups

$$\begin{aligned} \mathcal{E}(R(\vec{\theta}), \vec{0}) &\equiv R(\vec{\theta}) && \text{subgroup } O(m) \text{ of } E(m), \\ \mathcal{E}(\mathbb{K}, \vec{a}) &\equiv T(\vec{a}) && \text{subgroup } T(m) \text{ of } E(m) \quad (\text{translations in } m \text{ dimensions}). \end{aligned}$$

Comment: the subgroup $T(m)$ is normal. Take a pure translation $\mathcal{E}(\mathbb{I}, \vec{a})$ and a generic element $\mathcal{E}(R, \vec{b})$. We compute

$$\begin{aligned}\mathcal{E}(R, \vec{b}) \mathcal{E}(\mathbb{I}, \vec{a}) \mathcal{E}(R, \vec{b})^{-1} &= \mathcal{E}(R, \vec{b}) \mathcal{E}(\mathbb{I}, \vec{a}) \mathcal{E}(R^{-1}, -R^{-1}\vec{b}) \\ &= \mathcal{E}(R, \vec{b}) \mathcal{E}(R^{-1}, -R^{-1}\vec{b} + \vec{a}) = \mathcal{E}(RR^{-1}, -\vec{b} + R\vec{a} + \vec{b}) \\ &= \mathcal{E}(\mathbb{I}, R\vec{a}) \in T(m).\end{aligned}$$

Added explanation (not in the original notes)

What “normal” means and why it matters. A subgroup $N \subset G$ is *normal* if conjugating any of its elements by any group element lands you back in N : $gNg^{-1} = N$ for all $g \in G$. The computation above shows exactly that: conjugating a translation by any Euclidean transformation gives another translation (by the rotated vector $R\vec{a}$). Normality is the condition needed to form the quotient group $E(m)/T(m) \cong O(m)$ and, together with the factorization below, it is what makes $E(m)$ a *semi-direct* product. By contrast, $O(m)$ is *not* a normal subgroup of $E(m)$: conjugating a rotation by a translation gives a rotation about a shifted point, which is not a rotation about the origin.

2.5 Factorization and semi-direct product

[p. 8]

We have

$$\mathcal{E}[R(\vec{\theta}), \vec{a}] = \mathcal{E}(\mathbb{I}, \vec{a}) \mathcal{E}[R(\vec{\theta}), \vec{0}] = T(\vec{a}) R(\vec{\theta}).$$

It follows from the composition rule. *The order is important!* First we apply the orthogonal transformation, then the translation. We say that

$$E(m) = T(m) \rtimes O(m) \quad (\text{semi-direct product}).$$

Added explanation (not in the original notes)

Verify the factorization directly with the composition rule: $\mathcal{E}(\mathbb{I}, \vec{a}) \mathcal{E}(R, \vec{0}) = \mathcal{E}(\mathbb{I} \cdot R, \mathbb{I} \cdot \vec{0} + \vec{a}) = \mathcal{E}(R, \vec{a})$. ✓

“Semi-direct” ($\rtimes$) rather than “direct” ($\times$) because the two factors do not commute with each other: the composition rule mixes them through the twisted term $R_2\vec{a}_1$. Every element factors *uniquely* as (translation)·(orthogonal transformation), $T(m)$ is normal, and $T(m) \cap O(m) = \{e\}$ — these three facts are the definition of a semi-direct product.

2.6 Embedding of the Euclidean group in homogeneous coordinates

$E(m)$ is *not* a subgroup of $GL(m, \mathbb{R})$ since translations are not linear transformations. However, $E(m)$ is isomorphic to a subgroup of $GL(m+1, \mathbb{R})$. [p. 9] The isomorphism is realized by the map

$$\Phi : E(m) \rightarrow GL(m+1, \mathbb{R}), \quad \Phi(\mathcal{E}(R, \vec{a})) = \begin{pmatrix} R(\vec{\theta}) & \vec{a} \\ \vec{0}^T & 1 \end{pmatrix}$$

where $\vec{0}^T$ is a row vector of zeros.

■ The map preserves the group properties:

$$\Phi(\mathcal{E}(R_2, \vec{a}_2)) \Phi(\mathcal{E}(R_1, \vec{a}_1)) = \begin{pmatrix} R_2 & \vec{a}_2 \\ \vec{0}^T & 1 \end{pmatrix} \begin{pmatrix} R_1 & \vec{a}_1 \\ \vec{0}^T & 1 \end{pmatrix} = \begin{pmatrix} R_2 R_1 & R_2 \vec{a}_1 + \vec{a}_2 \\ \vec{0}^T & 1 \end{pmatrix} = \Phi(\mathcal{E}(R_2 R_1, R_2 \vec{a}_1 + \vec{a}_2)).$$

■ $\Phi(\mathcal{E}(R, \vec{a}))$ acts on $(m+1)$ -dimensional vectors of the form $\begin{pmatrix} \vec{x} \\ 1 \end{pmatrix}$ such that

$$\Phi(\mathcal{E}(R, \vec{a})) \begin{pmatrix} \vec{x} \\ 1 \end{pmatrix} = \begin{pmatrix} R & \vec{a} \\ \vec{0}^T & 1 \end{pmatrix} \begin{pmatrix} \vec{x} \\ 1 \end{pmatrix} = \begin{pmatrix} R\vec{x} + \vec{a} \\ 1 \end{pmatrix} \quad (\Phi \text{ acts linearly}).$$

↪ If we restrict to the first m entries, it realizes the transformation $\vec{x} \rightarrow R\vec{x} + \vec{a}$.

Added explanation (not in the original notes)

This is the “homogeneous coordinates” trick, ubiquitous in computer graphics and robotics: append a dummy coordinate equal to 1 to every point. The translation part $\vec{a}$ then sits in the last *column* of a bigger matrix and gets picked up by matrix multiplication against that trailing 1. The price is one extra dimension; the reward is that the whole of $E(m)$, translations included, now acts by honest matrix multiplication — so all the machinery of matrix Lie groups (exponential map, Lie algebra, etc.) becomes available. This is exactly what is exploited next.

3 The special Euclidean group $SE(m)$

[p. 10]

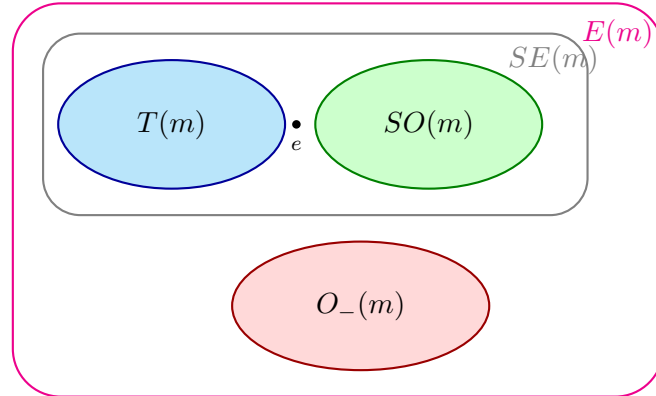
$O(m)$ is the disjoint union of $SO(m)$ and $O_-(m)$:

$$O(m) = SO(m) \sqcup O_-(m), \quad SO(m) \cap O_-(m) = \emptyset.$$

Since $RR^T = \mathbb{I} \rightsquigarrow (\det R)^2 = 1 \rightsquigarrow \det R = \pm 1$:

$$\begin{aligned} SO(m) &= \{R \in \mathbb{R}^{m \times m} \mid RR^T = R^T R = \mathbb{I} \wedge \det R = +1\}, \\ O_-(m) &= \{R \in \mathbb{R}^{m \times m} \mid RR^T = R^T R = \mathbb{I} \wedge \det R = -1\}. \end{aligned}$$

Consequently, the Euclidean group has the following structure:



$SE(m)$, the special Euclidean group, is the identity component of $E(m)$ and it consists of translations and special orthogonal transformations. [p. 11]

Added explanation (not in the original notes)

Why split off $O_-(m)$? Elements with $\det R = -1$ (reflections) cannot be reached continuously from the identity: $\det$ is a continuous function, so along any continuous path of orthogonal matrices it cannot jump from $+1$ to -1 . Hence $O(m)$ has (at least) two disconnected pieces, and only the piece containing $\mathbb{I}$ — namely $SO(m)$, and with translations $SE(m) = T(m) \rtimes SO(m)$ — can be written as exponentials e^{tX} of Lie-algebra elements. That is why the Lie-algebra analysis below is done for $SE(m)$ and not for the full $E(m)$. Note also

that $O_-(m)$ is *not* a subgroup: the product of two reflections has determinant $(+1)$ and is a rotation.

4 The Lie algebra of $SE(m)$

The Lie algebra of $SE(m)$ is the set of all matrices X such that $e^{tX} \in SE(m)$ for all $t \in \mathbb{R}$.

We use the embedding isomorphism: we look for an $(m+1)$ -dimensional matrix X such that

$$e^{tX} = \begin{pmatrix} R & \vec{a} \\ \vec{0}^T & 1 \end{pmatrix} \quad (\text{element of } SE(m), \text{ with } R \in SO(m)).$$

Consider

$$X = \begin{pmatrix} Y & \vec{v} \\ \vec{0}^T & 0 \end{pmatrix} \quad (Y: m \times m \text{ matrix}, \vec{v} \in \mathbb{R}^m)$$

and exponentiate:

$$X^2 = \begin{pmatrix} Y & \vec{v} \\ \vec{0}^T & 0 \end{pmatrix} \begin{pmatrix} Y & \vec{v} \\ \vec{0}^T & 0 \end{pmatrix} = \begin{pmatrix} Y^2 & Y\vec{v} \\ \vec{0}^T & 0 \end{pmatrix} \rightsquigarrow \boxed{X^k = \begin{pmatrix} Y^k & Y^{k-1}\vec{v} \\ \vec{0}^T & 0 \end{pmatrix}}$$

$$e^{tX} = \mathbb{I} + \sum_{k=1}^{\infty} \frac{t^k X^k}{k!} = \mathbb{I} + \sum_{k=1}^{\infty} \frac{t^k}{k!} \begin{pmatrix} Y^k & Y^{k-1}\vec{v} \\ \vec{0}^T & 0 \end{pmatrix}$$

[p. 12]

$$= \mathbb{I} + \begin{pmatrix} \sum_{k=1}^{\infty} \frac{t^k Y^k}{k!} & \sum_{k=1}^{\infty} \frac{t^k Y^{k-1}\vec{v}}{k!} \\ \vec{0}^T & 0 \end{pmatrix} = \begin{pmatrix} e^{tY} & \sum_{k=1}^{\infty} \frac{t^k Y^{k-1}\vec{v}}{k!} \\ \vec{0}^T & 1 \end{pmatrix} \stackrel{!}{\in} SE(m).$$

For this matrix to belong to $SE(m)$ we must have

$$e^{tY} \in SO(m) \rightsquigarrow Y^T = -Y,$$

and

$$\sum_{k=1}^{\infty} \frac{t^k Y^{k-1}\vec{v}}{k!} \in \mathbb{R}^m \rightsquigarrow \text{no specific requirements are imposed on } \vec{v} \in \mathbb{R}^m$$

(it is possible to show that this series indeed converges to some vector of $\mathbb{R}^m$).

$$\mathfrak{se}(m) = \left\{ X = \begin{pmatrix} Y & \vec{v} \\ \vec{0}^T & 0 \end{pmatrix} \mid Y^T = -Y \wedge \vec{v} \in \mathbb{R}^m \right\} \quad \text{Lie algebra of } SE(m).$$

Added explanation (not in the original notes)

Where does $Y^T = -Y$ come from? Demand $(e^{tY})^T e^{tY} = \mathbb{I}$ for all t . Differentiate at $t = 0$: $\frac{d}{dt} [e^{tY^T} e^{tY}]_{t=0} = Y^T + Y = 0$. Conversely, if $Y^T = -Y$ then $(e^{tY})^T = e^{tY^T} = e^{-tY} = (e^{tY})^{-1}$, so e^{tY} is orthogonal; and its determinant $e^{t \operatorname{tr} Y} = e^0 = 1$ (an antisymmetric matrix is traceless), so $e^{tY} \in SO(m)$ automatically — consistent with $SE(m)$ being the part of $E(m)$ connected to the identity.

Also, for invertible Y the translation series can be resummed explicitly: $\sum_{k \geq 1} \frac{t^k Y^{k-1}}{k!} \vec{v} = Y^{-1}(e^{tY} - \mathbb{I})\vec{v}$, which is manifestly a finite vector of $\mathbb{R}^m$.

5 Generators of $SE(m)$ for $m = 2$

[p. 13]

We look for a basis in $\mathfrak{se}(2)$.

First, we consider matrices in the form

$$X = \begin{pmatrix} Y & \vec{0} \\ \vec{0}^T & 0 \end{pmatrix}, \quad Y^T = -Y \quad \rightsquigarrow \quad \text{the most generic form of } Y \text{ is } Y = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}.$$

Therefore, a basis element is simply given by

$$T^3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{generator of rotation.}$$

Second, we consider elements of the form

$$X = \begin{pmatrix} 0 & \vec{v} \\ \vec{0}^T & 0 \end{pmatrix} \quad \rightsquigarrow \quad \text{since no restrictions are imposed on } \vec{v}, \text{ a basis is given by}$$

$$T^1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{generators of translations.}$$

[p. 14] By direct computation, we find

$$[T^1, T^2] = 0, \quad [T^3, T^1] = T^2, \quad [T^3, T^2] = -T^1.$$

Added explanation (not in the original notes)

Worked check of $[T^3, T^1] = T^2$: only the top-left 2×2 block of T^3 (call it Y_3) and the translation column of T^1 (call it $\vec{e}_1$) matter. Block multiplication gives $T^3 T^1 = \begin{pmatrix} 0 & Y_3 \vec{e}_1 \\ \vec{0}^T & 0 \end{pmatrix}$ and $T^1 T^3 = 0$ (its only nonzero column multiplies the zero bottom row of T^3). Since $Y_3 \vec{e}_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \vec{e}_2$, we get $[T^3, T^1] = T^2$. ✓ The other two follow the same way. Geometrically: rotating an infinitesimal translation along x gives a translation along y , and vice versa with a minus sign.

Two points:

(1) A generic element of the algebra X can be obtained as a *real linear combination of generators*. We write

$$X = \alpha_A T^A = -i\alpha_A (iT^A) = -i\alpha_A P^A, \quad P^A \equiv iT^A$$

where the P^A are the generators in the “convention of physicists”. By further defining $P^3 \equiv J$, we find

$$[P^1, P^2] = 0, \quad [J, P^1] = iP^2, \quad [J, P^2] = -iP^1.$$

[p. 15] **Obs.:** the P^i follow the same commutation of the momentum in QM, while J plays the role of the angular momentum.

Added explanation (not in the original notes)

The factor of i is pure convention. Mathematicians exponentiate real antisymmetric generators, e^{tT} ; physicists prefer *Hermitian* generators because Hermitian operators have real eigenvalues and correspond to observables. Since T^A is real and antisymmetric in the relevant block, $P^A = iT^A$ is Hermitian, and the group element is written $e^{-i\alpha_A P^A}$. This is exactly how momentum ($\hat{p} = -i\partial_x$, generator of translations) and angular momentum (generator of rotations) arise in ordinary quantum mechanics — the algebra we just derived from pure geometry *is* the algebra of QM observables. This is the “different interpretation of the same equations” announced on p. 2.

(2) A generic element of the *group* can be obtained by exponentiating an element of the algebra:

$$e^{-i\vec{a}\cdot\vec{P}} \in T(2), \quad e^{-i\theta J} \in SO(2).$$

As a final remark, we note that the algebra admits a Casimir operator, $|\vec{P}|^2$:

$$|\vec{P}|^2 \equiv (P^1)^2 + (P^2)^2, \quad \left[|\vec{P}|^2, P^i\right] = \left[|\vec{P}|^2, J\right] = 0.$$

Added explanation (not in the original notes)

What is a Casimir and why it matters. A Casimir operator commutes with *every* generator, hence with every group element in any representation. By Schur’s lemma, in an *irreducible* representation a Casimir must act as a multiple of the identity: a single number that labels the irrep. Here that number will be p^2 (the squared modulus of momentum) — compare with $\vec{J}^2 = j(j+1)$ labelling the spin- j irreps of $SU(2)$. Quick check that $\left[|\vec{P}|^2, J\right] = 0$: $[J, (P^1)^2] = P^1[J, P^1] + [J, P^1]P^1 = i(P^1P^2 + P^2P^1)$, and $[J, (P^2)^2] = -i(P^2P^1 + P^1P^2)$; they cancel in the sum. Physically: rotations do not change the length of the momentum vector.

6 Unitary irreps of $SE(2)$

Let us first recall a few basic definitions. [p. 16]

■ In group theory, a representation is a map that associates each element of the group with an invertible linear operator acting on a vector space, called the representation space.

■ The map is a homomorphism: it preserves the properties of the group.

$$D : G \rightarrow GL(V)$$

$$D(g_1)D(g_2) = D(g_1 \cdot g_2), \quad D(e) = \mathbb{I}_V, \quad D(g^{-1}) = D(g)^{-1},$$

where V is the representation space (a vector space over the field $\mathbb{K}$) and $GL(V)$ are the invertible linear operators acting on V . We use the notation

$$D[\mathcal{E}(R(\vec{\theta}), \vec{a})] = D(\vec{\theta}, \vec{a}).$$

Since the representation preserves the group properties, the factorization remains true in all representations. In the case $m = 2$, we write [p. 17]

$$D(\theta, \vec{a}) = D(\vec{a}) D(\theta) = e^{-i\vec{a} \cdot \vec{P}} e^{-i\theta J}.$$

The exponential map is always true; depending on the representation, $\vec{P}$ and J will take different forms.

■ The representation is irreducible (irrep) if there is no subspace of V left invariant under the action of G .

■ If the representation space is a complete inner product space (Hilbert space) and the $D(g)$ are unitary operators $\leadsto$ the representation is unitary (we will use $\mathcal{U}(g)$ instead of $D(g)$ in such case).

Added explanation (not in the original notes)

Why unitary irreps are what we want in QM. States live in a Hilbert space, and symmetry transformations must preserve transition probabilities $|\langle \phi | \psi \rangle|^2$ — by Wigner’s theorem this forces the symmetry to be implemented by (anti)unitary operators. *Irreducibility* is the group-theoretic notion of “elementary”: an irrep is a smallest set of states that transform only among themselves. The strategy of this whole course (following Wigner) is: *an elementary particle = a unitary irreducible representation of the spacetime symmetry group*. Here we practice on $SE(2)$; later the same logic applied to the Poincaré group will define relativistic particles and their mass and spin.

6.1 Construction of the unitary irreps of $SE(2)$

We focus on the case of $SE(2)$. We construct the unitary irreps as follows.

1) **We start from the algebra.** Since

$$[P^i, P^j] = \left[P^i, |\vec{P}|^2 \right] = 0,$$

we diagonalize these operators together: [p. 18]

$$\vec{P} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle, \quad |\vec{P}|^2 |\vec{p}\rangle = |\vec{p}|^2 |\vec{p}\rangle.$$

Since we are interested in unitary representations, the generators must be Hermitian and the eigenvalues real: $\vec{p} \in \mathbb{R}^2$. These equations are enough to characterize how the state $|\vec{p}\rangle$ transforms under the action of two-dimensional translations of $SE(2)$.

Added explanation (not in the original notes)

(The handwritten page says $\vec{p} \in \mathbb{R}^3$; for $SE(2)$ the momentum has two components, so it should read $\vec{p} \in \mathbb{R}^2$ — almost certainly a slip of the pen.)

If we write $\mathcal{U}(\vec{a}) = e^{-i\vec{a} \cdot \vec{P}}$, we get

$$\mathcal{U}(\vec{a}) |\vec{p}\rangle = e^{-i\vec{a} \cdot \vec{P}} |\vec{p}\rangle = e^{-i\vec{a} \cdot \vec{p}} |\vec{p}\rangle,$$

and the composition rule is satisfied:

$$\mathcal{U}(\vec{a}) \mathcal{U}(\vec{b}) |\vec{p}\rangle = \mathcal{U}(\vec{a}) e^{-i\vec{b} \cdot \vec{p}} |\vec{p}\rangle = e^{-i\vec{b} \cdot \vec{p}} \mathcal{U}(\vec{a}) |\vec{p}\rangle = e^{-i\vec{b} \cdot \vec{p}} e^{-i\vec{a} \cdot \vec{p}} |\vec{p}\rangle = e^{-i(\vec{a} + \vec{b}) \cdot \vec{p}} |\vec{p}\rangle = \mathcal{U}(\vec{a} + \vec{b}) |\vec{p}\rangle$$

$$\mathcal{U}(\vec{a}) \mathcal{U}(\vec{b}) = \mathcal{U}(\vec{a} + \vec{b}).$$

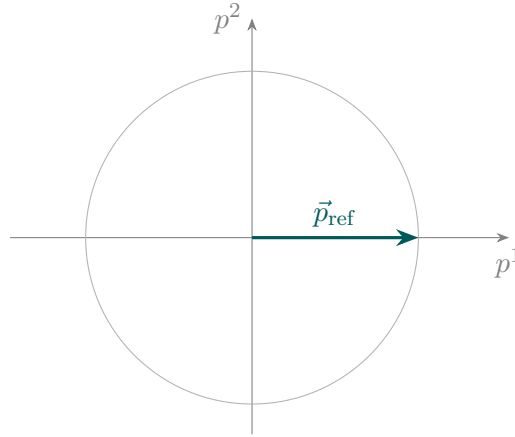
Added explanation (not in the original notes)

Note the step $e^{-i\vec{a}\cdot\vec{P}}|\vec{p}\rangle = e^{-i\vec{a}\cdot\vec{p}}|\vec{p}\rangle$: since $|\vec{p}\rangle$ is an eigenstate of every component of $\vec{P}$, any function of $\vec{P}$ acting on it can be replaced by the same function of the eigenvalue $\vec{p}$ (expand the exponential in a power series to see it). On its eigenstates a translation therefore acts just as a *phase* — exactly like a plane wave $e^{i\vec{p}\cdot\vec{x}}$ picks up the phase $e^{i\vec{p}\cdot\vec{a}}$ under $\vec{x} \rightarrow \vec{x} + \vec{a}$.

2) We need to understand how rotations act on the states $|\vec{p}\rangle$. [p. 19] $\hookrightarrow$ since $[J, P^i] \neq 0$, we have $J|\vec{p}\rangle \neq |\vec{p}\rangle$! We take the following steps.

2a) The vector $\vec{p}$ is described by a modulus $|\vec{p}| \equiv p$ and a direction in space $\hat{p}$ that we know once the components of $\vec{p}$ are specified. We fix the modulus $p = |\vec{p}| > 0$ (related to the eigenvalue of the Casimir operator $|\vec{P}|^2$ via $|\vec{P}|^2|\vec{p}\rangle = p^2|\vec{p}\rangle$) and select a “*standard reference vector*” $\vec{p}_{\text{ref}}$ with modulus p and fixed direction in the plane. For instance,

$$\vec{p}_{\text{ref}} = \begin{pmatrix} p \\ 0 \end{pmatrix} \quad \text{and the eigenvalue equations are} \quad \begin{cases} P^1 |\vec{p}_{\text{ref}}\rangle = p |\vec{p}_{\text{ref}}\rangle \\ P^2 |\vec{p}_{\text{ref}}\rangle = 0 \\ |\vec{P}|^2 |\vec{p}_{\text{ref}}\rangle = p^2 |\vec{p}_{\text{ref}}\rangle \end{cases}$$



2b) What happens if we act on this state with a rotation? [p. 20] Given $\mathcal{U}(\theta) = e^{-i\theta J}$, what is $\mathcal{U}(\theta)|\vec{p}_{\text{ref}}\rangle$?

To answer this question, we act on the above state with P^k . We write

$$P^k \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle = \mathcal{U}(\theta) \underbrace{[\mathcal{U}(\theta)^{-1} P^k \mathcal{U}(\theta)]}_{\text{what is this string of operators?}} |\vec{p}_{\text{ref}}\rangle \stackrel{*}{=}$$

To gain some insight, we consider a generic representation and the combination of transformations $D(\theta)^{-1}D(\vec{a})D(\theta)$. We take $\vec{a}$ *infinitesimal* and elaborate as follows. Since $\vec{a}$ is infinitesimal, we expand:

$$D(\theta)^{-1}D(\vec{a})D(\theta) = D(\theta)^{-1}(\mathbb{I} - i\vec{a} \cdot \vec{P})D(\theta) = \mathbb{I} - i a^k D(\theta)^{-1} P^k D(\theta).$$

We can also elaborate with the composition rule first, and then expand:

$$D(\theta)^{-1}D(\vec{a})D(\theta) = D(\theta)^{-1}D(\theta, \vec{a}) = D(R(\theta)^{-1}\vec{a}) = \mathbb{I} - i(R(\theta)^{-1}\vec{a})^j P^j$$

(still a translation, but by the vector $R(\theta)^{-1}\vec{a}$). [p. 21]

$$\mathbb{I} - i(R(\theta)^{-1})_{jk} a^k P^j = \mathbb{I} - i(R(\theta))_{kj} a^k P^j \quad \text{since } RR^T = \mathbb{I} \rightsquigarrow R^{-1} = R^T.$$

From the comparison, we get

$$D(\theta)^{-1} P^k D(\theta) = (R(\theta))_{kj} P^j \quad \text{valid in all representations,}$$

where $R(\theta)$ is the 2×2 matrix that represents the rotation in $\mathbb{R}^2$.

Added explanation (not in the original notes)

Reading this identity. It says that the abstract operators P^k transform *as the components of a vector* under conjugation by a rotation. The two computations compare the same object in two ways: (i) sandwiching the infinitesimal translation between $D(\theta)^{-1}$ and $D(\theta)$; (ii) using the group composition rule $\mathcal{E}(R_2, \vec{a}_2) \mathcal{E}(R_1, \vec{a}_1) = \mathcal{E}(R_2 R_1, R_2 \vec{a}_1 + \vec{a}_2)$ to show that this sandwich is itself a pure translation by the rotated vector $R(\theta)^{-1} \vec{a}$. Matching the coefficients of the arbitrary infinitesimal a^k gives the boxed identity. Also used above: for a rotation matrix $(R^{-1})_{jk} = (R^T)_{jk} = R_{kj}$.

Back to $\otimes$, we write

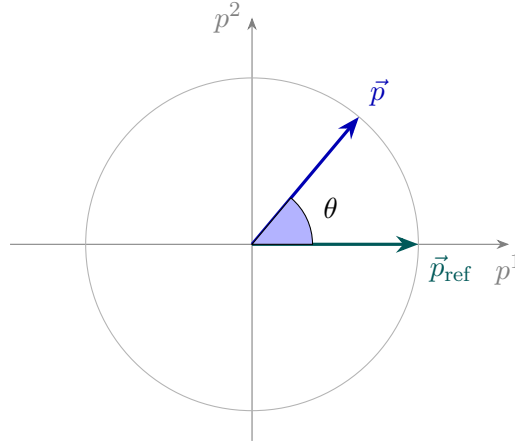
$$\stackrel{\otimes}{=} \mathcal{U}(\theta) (R(\theta))_{kj} P^j |\vec{p}_{\text{ref}}\rangle = \mathcal{U}(\theta) \underbrace{(R(\theta))_{kj} p_{\text{ref}}^j}_{\text{we introduce } \vec{p} \equiv R(\theta) \vec{p}_{\text{ref}}} |\vec{p}_{\text{ref}}\rangle = \mathcal{U}(\theta) p^k |\vec{p}_{\text{ref}}\rangle = p^k \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle,$$

where we used $P^j |\vec{p}_{\text{ref}}\rangle = p_{\text{ref}}^j |\vec{p}_{\text{ref}}\rangle$. We thus get

$$P^k \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle = p^k \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle$$

$\hookrightarrow \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle$ is an eigenstate of $\vec{P}$ with eigenvalue $\vec{p} = R(\theta) \vec{p}_{\text{ref}}$. We thus define [p. 22]

$$|\vec{p}\rangle \equiv \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle \quad (\star) \quad \text{with} \quad \vec{P} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle, \quad \left| \vec{P} \right|^2 |\vec{p}\rangle = p^2 |\vec{p}\rangle.$$



3) We now consider the set of vectors

$$\left\{ |\vec{p}\rangle = \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle, \quad \theta \in [0, 2\pi) \right\}.$$

The claim is that these states span the representation space of a unitary irrep of $SE(2)$.

N.B.: all these states are eigenstates of $\left| \vec{P} \right|^2$ with the same eigenvalue p^2 fixed (that is, they all have the same energy). We already know how translations act on this states...

Added explanation (not in the original notes)

Summary of the construction and what comes next. For each fixed $p > 0$ (the Casimir label) we built a family of states $\{|\vec{p}\rangle\}$, one for every point of the circle of radius p in momentum space:

- translations act diagonally: $\mathcal{U}(\vec{a})|\vec{p}\rangle = e^{-i\vec{a}\cdot\vec{p}}|\vec{p}\rangle$;
- rotations move states around the circle: $\mathcal{U}(\theta')|\vec{p}\rangle = |R(\theta')\vec{p}\rangle$, which follows from $(\star)$ and the group law $\mathcal{U}(\theta')\mathcal{U}(\theta) = \mathcal{U}(\theta' + \theta)$.

No smaller subset of the circle is left invariant by all rotations, which is why the representation is *irreducible*: starting from any single state you can reach all the others. Note it is infinite-dimensional — a general feature of unitary irreps of non-compact groups such as $SE(2)$ (and, later, the Poincaré group). The analogy to keep in mind for the relativistic case: circle of radius $p \leftrightarrow$ mass-shell hyperboloid $p^\mu p_\mu = m^2$; reference vector $\vec{p}_{\text{ref}} \leftrightarrow$ rest-frame momentum $(m, \vec{0})$; and the rotations that move $\vec{p}_{\text{ref}}$ around $\leftrightarrow$ Lorentz boosts/rotations. This is Wigner's method of induced representations, of which today's lecture is the simplest example.

(End of Lecture 1 — the notes continue with how rotations and the full group act on this basis.)